

AR 523: Robot Manipulators

Course Code: AR 523

Course Name: Robot Manipulators

L-T-P-C: 2-1-0-3

Prerequisites: IC 114: Linear Algebra / Introductory course in Linear Algebra, ME 206: Mechanics of Solids / ME 209: Dynamics, EE 301: Control Systems, Programming Knowledge (Python)

Intended for: B.Tech, M.Tech, M.Tech. (R), PhD

Mutual Exclusion: None

Approval: 48th Senate (listed in the B.Tech Quantum Technology curriculum, 3-0-0-3); 64th BoA (recoded from AR 501: Robot Kinematics, Dynamics and Control; L-T-P-C 2-1-0-3)

Preamble

Robot manipulators are one of the most widely used and studied classes of robots in both industrial and research settings. These robotic arms are designed to interact with the environment through physical manipulation and are essential in fields ranging from manufacturing and logistics to medical robotics and space exploration. The study of robot manipulators lies at the intersection of mechanical systems, control theory, and computer science, and provides a foundation for understanding broader robotic systems.

In this course, students will explore the modeling, analysis, and control of robot manipulators. The course will build mathematical foundations using concepts from linear algebra, differential equations, rigid body mechanics, and system dynamics. Students will also study classic and modern control techniques. Emphasis will be placed on practical applications through programming assignments and lab work, with tools such as Python and ROS forming a core part of the learning.

Course Modules

- **Introduction:** Basic definitions, mechanism, degree of freedom, Rigid Body Motions, Serial and parallel manipulators (4 hours)
- **Kinematics:** Kinematic parameters, Denavit-Hartenberg (DH) representation, Forward and inverse kinematics, Analytical and numerical solutions, Differential kinematics, Jacobian matrix, Mapping between configuration-space to operational-space, Pseudo inverse concepts, Introduction to workspace singularities (10 hours)
- **Dynamics:** Forward and inverse dynamics. Lagrangian (Lagrange-Euler) and Newton-Euler formulations. Dynamic simulation: Dynamic modeling of robotic manipulators and computer-based numerical simulations (10 hours)
- **Motion Planning:** Path and Trajectory planning, Configuration (joint) space trajectory and operational (task) space trajectory generations (8 hours)
- **Control:** Velocity control, Joint space and task-space control schemes, Force control: Direct, Indirect, Hybrid position/force, Vision-based control, Advanced control: Reinforcement learning, Imitation learning (10 hours)

Text Books

1. *Modern Robotics: Mechanics, Planning, and Control* by K.M. Lynch and F.C. Park, Cambridge University Press, 2017.
2. *Introduction to Robotics*, Third Edition, by S. K. Saha, McGrawHill, 2024.

References

1. *A Mathematical Introduction to Robotic Manipulation* by Murray et al, CRC Press, 1994.
2. *Fundamentals of Robotic Mechanical Systems* by J. Angeles, Springer Verlag, 200
3. *Robotics: Modelling, Planning and Control* by B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, Springer, 2009.
4. *Introduction to Robotics: Mechanics and Control* by J.J. Craig, Pearson, 4th Edition, 2017.
5. *Robot Modeling and Control* by M.W. Spong, S. Hutchinson, and M. Vidyasagar, Wiley, 2006.
6. *Principles of Robot Motion: Theory, Algorithms, and Implementations* by H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, L.E. Kavraki, and S. Thrun, MIT Press, 2005.